LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.5.3 – Device Interfaces: Display

Lesson Overview:

In this lesson, students will compare different input and output interfaces for display devices.

Students will:

- Compare input and output interfaces for display devices.
- Analyze scenarios and determine which display interface best meets the user's needs.

Guiding Question: What are the different input and output interfaces for display devices and how do they work?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 45-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.5 – Compare and contrast common types of input/output device interfaces.
  - Display ports
    - Video Graphics Array (VGA)
    - Digital Visual Interface (DVI)
    - High Definition Media Interface (HDMI)
    - DisplayPort
    - USB-C
  - Display technology
    - Mirroring
    - Casting

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Lesson Background

Background Information:

Display interfaces are the connections and standards that carry video and audio signals to a screen, while display technologies determine how that content is shown on the screen. Video graphics array (VGA) and digital visual interface (DVI) both support analog signals, but DVI can also carry digital signals and typically offers better image quality than VGA. High-Definition Media Interface and DisplayPort both support video and audio through digital signals at high resolution and refresh rates. USB-C is being widely adapted in modern technology and can support audio, video, and power supply. Mirroring and casting are two technologies that allow a user to duplicate or extend their display to other devices.

By the end of this lesson, students should be able to evaluate a scenario and determine which display interface is needed.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.5.3 – Device Interfaces Display • Activity Slides 2.5.3 – Display Interface Recommendation
Materials per Student	• Student Notes 2.5.3 – Device Interfaces Display • Student Handout 2.5.3 – Device Interfaces Display • Assessment 2.5.3 – Device Interfaces Display (optional)

Lesson Background

Cyber Terms:

Term	Definition
display interfaces	connections and standards that allow devices to send video and audio signals to a display
Video Graphics Array (VGA)	an analog video standard widely used for connecting monitors and projectors to computers
Digital Visual Interface (DVI)	a digital video interface that can carry both analog and digital signals, used for connecting monitors to graphic cards
High-Definition Media Interface (HDMI)	a digital interface that supports both video and audio signals, commonly used in electronics like TVs, monitors, and laptops
DisplayPort	a digital interface designed for high-performance display connection, used in computer monitors and graphics cards
USB-C	a versatile, reversible connector that can handle various types of data, including video, audio, and power
mirroring	the process of duplicating the display of one device onto another
casting	the process of sending specific content from one device to another while the source device handles processing and the target display shows the content

2.5.3 – Device Interfaces: Display

Guiding Question: What are the different input and output interfaces for display devices and how do they work?

Lesson Launch (5 minutes)

- Display interfaces are the connections and standards that allow devices to send video and audio signals to a display.
- Display technology refers to how visual content is shown on a screen.
- Display interfaces and technologies are important because:
  - They enhance visual experience.
  - They boost productivity with high resolution and refresh rates.
  - They provide flexibility and compatibility across devices.

Display Interfaces (15-20 minutes)

- Distribute Student Handout 2.5.3 – Device Interfaces: Display and encourage students to fill in the answers during the lesson.
- Introduce ***Video Graphics Array (VGA)***, its definition, and features.
- Introduce ***Digital Visual Interface (DVI)***, its definition, and features.
- Introduce ***High-Definition Media Interface (HDMI)***, its definition, and features.
  - CEC is an HDMI standard that allows multiple HDMI connected devices to communicate and be controlled with a single remote control.
  - Version Example: HDMI 1.0 has basic high-definition video and audio while HDMI 2.0 has the bandwidth for 4K at 60Hz.
- Introduce ***DisplayPort***, its definition, and features.
- Introduce ***USB-C***, its definition, and features.
- Introduce ***Mirroring***, its definition, and features.
  - Mirroring Technology:
    - Apple AirPlay Mirroring: used with Apple devices to mirror screens to Apple TV or compatible devices.
    - Miracast: A standard for wireless screen mirroring used in Windows and Android devices to connect compatible receivers.
    - Google Chromecast (Mirror Mode): Mirrors a computer or a mobile device.
- Introduce ***Casting***, its definition, and features.
  - Casting Technology:

- Google Chromecast: Allows the casting of videos, music, and other content from apps to a TV or monitor.
- Apple AirPlay: Allows casting of media content from Apple devices to Apple TV or compatible smart TVs.
- Digital Living Network Alliance (DLNA): A standard for sharing media content over a network to compatible devices.

Activity: Display Interface Recommendation (20-25 minutes)

- Display Activity Slides 2.5.3 – Display Interface Recommendation, have students write their answers for each scenario at the bottom of the student handout.
- Choose if you want students to work together or in a small group.
- Read each scenario aloud or let students read independently or in their group.
- Students will determine which display interface meets the user's needs.
- Provide students with time to write their answers.

Scenarios and Answers

- Jordan is a freelancer who edits 4k videos on his laptop. He needs to connect an external monitor, an SSD, and keep his laptop charged for long work sessions. Which interface would handle these tasks effectively?

  Thunderbolt is the best choice of interface because it can support data transfer, high-resolution video output, and power delivery through one cable.

- Jess is setting up a presentation on an older projector that only supports analog video. Which display interface should she use?

  Jess should use the VGA interface because it is the only interface that supports analog signals for video display.

- Sallie wants to connect two monitors to a gaming PC for high refresh rate gaming. Which interface is best suited for Sallie's needs?

  Sallie should use a DisplayPort because it is designed for high-performance display connection and supports high resolution and refresh rates.

- Cindy wants to play a Blu-Ray movie with full HD and audio on her Smart TV. Which display interface should Cindy use?

  Cindy should use the HDMI interface because it supports both video and audio signals and will allow her to watch a movie with one cable.

- Linda has a thin, modern laptop and she needs to connect it to an external 4K monitor with built-in speakers. She also wants the cable to charge her laptop and send video and audio to other displays. Which display interface should Linda use?

  Linda should use the USB-C interface because it is versatile and allows users to handle various types of data, including video, audio, and power.

- Jeff's office PC supports the analog video standard, and he wants the highest image quality possible for his computer. Which display interface should he use?

  Jeff should choose the DVI interface because it is compatible with analog signals and has a better image quality than VGA.

- Karlie needs to show her laptop screen on a big TV during a meeting while still having control from her laptop. What display technology would allow her to do this?

  Karlie should use mirroring because it will allow her to duplicate her device onto another device and have control from her source device.

- Austin wants to play a video from his phone on his TV, while still using his phone for other tasks. What display technology will allow him to do this?

  Austin should use casting because it will allow him to send the video from his phone to his TV and still use his phone for other tasks without interrupting the video on the TV display.

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What are display interfaces and technologies important?

  Display interfaces and technologies are important because they enhance visual experience, boost productivity with high resolution and refresh rates, and they provide flexibility and compatibility across devices.

- What should a user consider when choosing a display interface port?

  A user should consider what type of device they have, if their device carries analog or digital signals, what quality of video they would like, and if they need to display audio, video, or both.

- What is the difference between mirroring and casting?

  Mirroring duplicates the source device to the target device, while casting extends the source device to the target device and allows the source device to run and complete other tasks simultaneously.